Pharmacy Query Assistant: Open-Weight vs Closed LLM Evaluation

Literature review supporting the Pharmacy Query Assistant project

1. Introduction: Generative AI and Large Language Models in Healthcare

Large language models (LLMs) are transformer-based neural networks [1] pre-trained on very large text corpora and then aligned to follow instructions and human preferences [2]. Their rapid uptake in medicine began when general-purpose models showed that clinical knowledge can emerge from scale. GPT-4 exceeded the USMLE passing threshold by a wide margin without medical fine-tuning [3], and ChatGPT performed at or near the passing level on the same examination [4]. Med-PaLM showed that instruction prompt tuning could raise accuracy on multiple-choice medical benchmarks and proposed a human evaluation framework covering factuality, harm and bias [5]. Med-PaLM 2 later reached 86.5% on MedQA [6].

Surveys describe wide potential: documentation support, patient communication, education and clinical decision support [8], [9], [10]. Early evidence on patient communication was encouraging. In a cross-sectional study, evaluators preferred chatbot answers over physician answers to online patient questions in 78.6% of evaluations, rating them higher for quality and empathy [11]. Yet the same literature stresses that fluent output is not the same as reliable output. LLMs can produce confident but unsupported statements, a failure mode known as hallucination [12], [13], and may reproduce biased or outdated medical beliefs [14], [15]. Lee et al. [7] argue that these limits make human oversight and careful evaluation essential. Pharmacy is a particularly unforgiving setting for such errors, because a wrong statement about dose, interaction or contraindication can directly cause patient harm.

2. Existing AI-Powered Pharmacy Assistants and Medical Chatbots

2.1 From rule-based agents to generative systems

Earlier healthcare chatbots were mostly rule-based or retrieval-based and covered narrow tasks such as symptom triage, adherence reminders and appointment booking. Systematic reviews found promising but low-quality evidence, with small samples, short follow-up and inconsistent outcome measures [16], [17]. Generative LLMs removed the need for hand-written dialogue trees but introduced new risks of open-ended error.

2.2 General medical LLM assistants

Several lines of work now exist. Proprietary systems include Med-PaLM and Med-PaLM 2 [5], [6] and the research prototype AMIE, a conversational diagnostic agent evaluated in simulated consultations [18]. Open-source medical assistants include ChatDoctor, a LLaMA model fine-tuned on doctor–patient dialogues and augmented with online retrieval [19], and MedAlpaca [20]. These systems are mostly designed for diagnosis, clinical dialogue or exam-style questions rather than medication information.

2.3 Pharmacy-focused evaluations

Work specifically on pharmacy is smaller and largely consists of evaluations of general chatbots rather than purpose-built assistants. Al-Ashwal et al. [21] tested ChatGPT-3.5, ChatGPT-4, Bing AI and Bard on drug–drug interaction queries against conventional interaction checkers and found variable sensitivity, specificity and accuracy across tools. Huang, X. et al. [22] compared ChatGPT with clinical pharmacists across pharmacy tasks and found the model reasonable on drug-information queries but weaker on more complex tasks such as medication consultation and case analysis. Roosan et al. [23] examined ChatGPT for medication therapy management and similarly emphasised the need for professional verification. Together these studies show potential for routine drug information but a lack of validated, deployable pharmacy assistants.

3. Open-Weight versus Closed LLMs

3.1 Architectures

Both families use decoder-only transformer architectures trained for next-token prediction and then aligned through instruction tuning and preference optimisation [1], [2]. The difference lies mainly in access, not design. Closed models such as GPT-4 are reachable only through an API, and their architecture, training data and parameter counts are not disclosed [24]. Open-weight models such as Llama 2, Llama 3, Mistral 7B and Gemma publish their weights, and often technical reports, so they can be downloaded, inspected and run locally [25], [26], [27], [28]. "Open-weight" is not always "open-source": licences may restrict use, and training data are usually not released.

3.2 Advantages and limitations

Dimension	Closed (API) LLMs	Open-weight LLMs
Raw capability	Generally highest on medical benchmarks [3], [6]	Improving quickly; smaller models lag on complex reasoning [26]
Privacy / data control	Data sent to a third party; compliance burden	Can be run on-premises, so sensitive queries stay local
Customisation	Limited fine-tuning options	Full fine-tuning or parameter-efficient adaptation such as LoRA/QLoRA [29], [30]
Cost	Per-token pricing; cost scales with usage [31]	Hardware and engineering cost; low marginal cost per query
Reproducibility	Models may be silently updated or deprecated	Fixed weights allow reproducible evaluation
Transparency	Opaque training data and safeguards	Inspectable, though training data are often undisclosed
Safety tuning	Vendor-managed guardrails	Responsibility shifts to the deployer

3.3 Medical adaptation of open models

Domain-adapted open models show that the gap can be narrowed. MEDITRON-70B, continually pre-trained on clinical guidelines and PubMed, outperformed its Llama 2 base on medical benchmarks [32]. PMC-LLaMA, pre-trained on biomedical papers and books, improved over general LLaMA models on medical question answering [33]. Even so, such models are rarely evaluated on drug-specific questions, and fine-tuning does not by itself remove hallucination.

4. Applications of LLMs in Pharmaceutical Question Answering

Pharmaceutical question answering (QA) covers indications, dosing, contraindications, adverse effects, interactions, counselling and prescription checking. Existing medical QA benchmarks such as MedQA [34], MedMCQA [35] and PubMedQA [36] include pharmacology content but are exam-style or research-abstract tasks and do not mirror real pharmacy queries. Three approaches appear in the literature.

Prompting a general model. Studies such as Al-Ashwal et al. [21], Huang, X. et al. [22] and Roosan et al. [23] query chatbots directly. Results are mixed and depend on question complexity, with no standard protocol.

Fine-tuning. Instruction tuning on medical text, as in ChatDoctor, MedAlpaca and MEDITRON, injects domain language and style [19], [20], [32]. Knowledge, however, is frozen at training time, which is a problem for drug information that changes with new warnings and approvals.

Retrieval-augmented generation (RAG). RAG conditions the model on retrieved passages [38], [39]. In medicine, Almanac used curated clinical sources and improved factuality, completeness and safety relative to standard LLMs in physician ratings [40]. Xiong et al. [37] benchmarked RAG across medical QA tasks and reported accuracy gains of up to about 18% over chain-of-thought prompting alone, while also noting that retrieval quality and positional effects matter, consistent with the "lost in the middle" problem [41]. For pharmacy, authoritative structured sources such as DrugBank [42] and regulatory labels are natural knowledge bases.

5. Evaluation Methods for Medical LLMs

A systematic review found that most evaluations of healthcare LLMs relied on exam-style questions or synthetic cases, with few using real patient-care data, and that assessment dimensions were inconsistent [43]. The following dimensions are most relevant to a pharmacy assistant.

Accuracy. Benchmark accuracy on multiple-choice sets [34], [35] is easy to compute but weakly predictive of open-ended answer quality. For free text, expert grading against a reference is the standard, with accuracy, sensitivity and specificity used in interaction studies [21].

Relevance and quality. Singhal et al. [5] introduced rater rubrics covering scientific consensus, comprehension, reasoning, possible harm and bias. Holistic frameworks such as HELM assess multiple metrics together [44]. Retrieval-based systems can additionally use faithfulness and answer-relevance metrics [45].

Hallucination. Med-HALT tests hallucination in medical tasks and found that both proprietary and open models performed poorly on several hallucination tests [46]. Surveys catalogue types and mitigation methods [12], [13]. Practical measures include citation or source grounding checks and unsupported-claim rates.

Safety and bias. Safety evaluation covers potential for harm, incorrect dosing, missed contraindications and biased content [5], [14]. Regulatory analysis suggests chatbots giving medical advice may qualify as medical devices [47].

Response time and cost. These are far less reported than accuracy. Chen, L. et al. [31] show that API costs vary by orders of magnitude between models and that routing can sharply reduce cost without lowering quality. For local deployment, latency and throughput depend on model size and quantization [30].

6. Comparison of Existing Approaches and Their Findings

Study	Task / domain	Approach	Key finding	Gap relevant to pharmacy
Singhal et al. [5]	Medical QA	Instruction prompt-tuned Flan-PaLM / Med-PaLM (closed)	Strong benchmark accuracy; human rubric exposed remaining gaps	Not pharmacy-specific; not publicly accessible
Singhal et al. [6]	Medical QA	Med-PaLM 2	86.5% on MedQA	Closed; no cost/latency analysis
Nori et al. [3]	Medical challenge problems	GPT-4, zero/few-shot	Exceeded USMLE pass mark with wide margin	Exam questions only
Ayers et al. [11]	Patient questions	ChatGPT vs physicians	Chatbot preferred in 78.6% of ratings	Social-media questions; no drug focus
Al-Ashwal et al. [21]	Drug–drug interactions	ChatGPT-3.5/4, Bing, Bard vs interaction checkers	Performance varied; chatbots not equivalent to conventional tools	Closed models; no grounding
Huang, X. et al. [22]	Clinical pharmacy tasks	ChatGPT vs pharmacists	Adequate on drug information, weaker on complex tasks	Single closed model; no retrieval
Roosan et al. [23]	Medication therapy management	ChatGPT	Useful potential; verification required	Limited scope
Li et al. [19]	Patient dialogue	ChatDoctor (fine-tuned LLaMA + retrieval)	Better than base LLaMA/ChatGPT on its metrics	Not drug-focused
Chen, Z. et al. [32]	Medical QA	MEDITRON (open, continued pre-training)	Outperformed Llama 2 base on medical benchmarks	Benchmarks, not pharmacy queries
Zakka et al. [40]	Clinical QA	Almanac (RAG)	Improved factuality, completeness, safety	Limited to clinical scenarios, closed backbone
Xiong et al. [37]	Medical QA	RAG benchmark (MIRAGE)	Accuracy gains up to ~18% over CoT	General medical; cost and latency secondary
Pal et al. [46]	Hallucination	Med-HALT	Hallucination widespread in proprietary and open models	Not drug-specific

Three patterns emerge. First, closed models lead in raw accuracy, but open and domain-adapted models are competitive on selected tasks. Second, grounding through RAG consistently improves factuality across studies. Third, comparisons are difficult because datasets, rubrics and raters differ.

7. Limitations of Current Systems and Research Gaps

1. **Lack of pharmacy-specific benchmarks.** Exam datasets dominate, and pharmacy studies use small, ad hoc question sets [43], [22].
2. **Reliance on closed models.** Most pharmacy studies test ChatGPT-style systems, limiting reproducibility, privacy control and customisation [21], [22].
3. **Insufficient grounding in authoritative drug sources.** Direct prompting yields unreliable interaction and dosing information, while RAG studies target general clinical questions [40], [37].
4. **Hallucination and safety.** Hallucinations persist even in strong models [46], [12], and few studies measure harm explicitly for medication advice.
5. **Unreported efficiency.** Response time and cost are rarely compared across open and closed models, although they determine real-world feasibility [31].
6. **Inconsistent evaluation.** Metrics and protocols are not standardised, and human grading is costly [43], [48].
7. **Bias, privacy and regulation.** Bias propagation and unclear regulatory status remain unresolved [14], [47].

8. How the Proposed Pharmacy Query Assistant Addresses These Gaps

The Pharmacy Query Assistant is positioned as a retrieval-grounded question-answering system for pharmacy queries that compares open-weight and closed LLMs under a single protocol. Its design responds to the gaps above as follows.

Research gap	Response in the Pharmacy Query Assistant
Few pharmacy-specific benchmarks	A dedicated set of pharmacy queries (dosing, interactions, contraindications, adverse effects, counselling) reviewed by domain experts
Dependence on closed models	Side-by-side evaluation of open-weight and closed LLMs, enabling local, privacy-preserving deployment where open models suffice
Weak grounding	RAG over authoritative drug references (e.g., DrugBank-style databases and product labels) so answers can cite sources [38], [42]
Hallucination and safety	Source-faithfulness checks, explicit unsupported-claim and harm ratings, and refusal or referral to a pharmacist when evidence is lacking [46], [5]
Unreported cost and latency	Measured response time and per-query cost alongside accuracy and relevance [31]
Inconsistent evaluation	A unified rubric covering accuracy, relevance, hallucination, safety, response time and cost, with optional calibrated LLM-as-judge scoring [48], [45]

By combining grounded retrieval, a multi-dimensional evaluation protocol and a direct open-versus-closed comparison in a pharmacy setting, the project addresses areas the current literature leaves open. It should be presented as decision support for pharmacists and not as a replacement for professional judgement.

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